



PMIT-6121: MACHINE LEARNING

JAHANGIRNAGAR UNIVERSITY

INSTITUTE OF INFORMATION TECHNOLOGY (IIT)

Project: Comparative Analysis of Machine Learning Classifiers for Heart Disease Prediction

Authors:

Asir Ahbab Raiyan (ID: 252011)

Date: June 14, 2026

Abstract

Heart disease is a leading cause of death worldwide and early prediction is crucial to improve healthcare outcomes. In this work, we are using machine learning (ML) techniques to predict heart disease with the help of the Personal Key Indicators of the heart disease dataset. The data set was preprocessed by removing duplicates, encoding categorical features, splitting into training and test sets, and scaling the features. Exploratory data analysis (EDA) was performed to study the distribution of features, class imbalance, and the relationship between variables. Three classification algorithms were implemented and compared: Support Vector Machine (SVM), K-Nearest Neighbors (KNN) and Gaussian Naive Bayes (GNB). Several SVM kernels and K values for KNN were tested to improve performance. The results showed that Gaussian Naive Bayes offered a competitive prediction accuracy in a low computational complexity. In general, the study demonstrates the potential of machine learning models to support early assessment of heart disease risk and assist healthcare professionals in decision-making processes.

1 Introduction & Exploratory Data Analysis

1.1 Introduction

Heart disease remains one of the most serious public health challenges worldwide. Early detection of cardiovascular diseases can significantly reduce mortality rates and improve treatment outcomes. Traditional diagnosis often requires multiple clinical tests and expert evaluation. Machine learning techniques provide an opportunity to analyze patient health indicators and identify individuals at risk of developing heart disease in a timely manner. The objective of this project is to develop and compare multiple machine learning models capable of predicting the probability of heart disease based on personal health indicators.

1.2 Dataset Overview

The dataset used in this study is the Personal Key Indicators of Heart Disease dataset obtained from the CDC Behavioral Risk Factor Surveillance System. The characteristics of Datasets are mentioned below:

1. Total Records: 319,795
2. Total Features: 18
3. Target Variable: HeartDisease
4. Total Records: 319,795

5. Positive Cases (Heart Disease = Yes): 27,373

6. Negative Cases (Heart Disease = No): 292,422

The target variable exhibited a significant class imbalance, with approximately 8.56% positive cases and 91.44% negative cases.

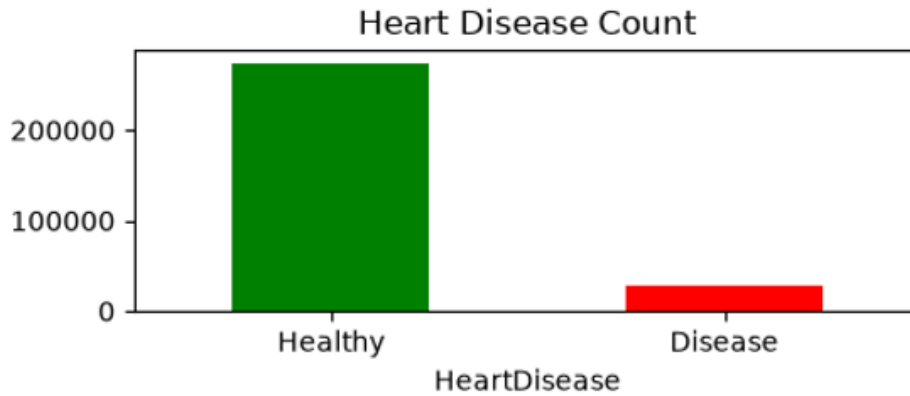


Figure 1: Heart Disease Distribution

1.3 Correlation Analysis

A correlation heatmap was generated after converting categorical variables into numerical representations using one-hot encoding. The analysis revealed that variables such as Age Category, General Health, Difficulty Walking, Stroke History and Physical Health showed relatively stronger relationships with heart disease occurrence.

2 Methodology & Data Preprocessing

2.1 Data Cleaning

The following preprocessing steps were performed:

1. At first, we loaded the dataset into a Pandas DataFrame.
2. Then, we verified the absence of missing values. There were no missing values in the dataset which is indeed a good sign.
3. After that, we checked for duplicate records and removed them for better training of the models.
4. Lastly, we converted the target variable (Heart Disease): Yes $\rightarrow$ 1, No $\rightarrow$ 0

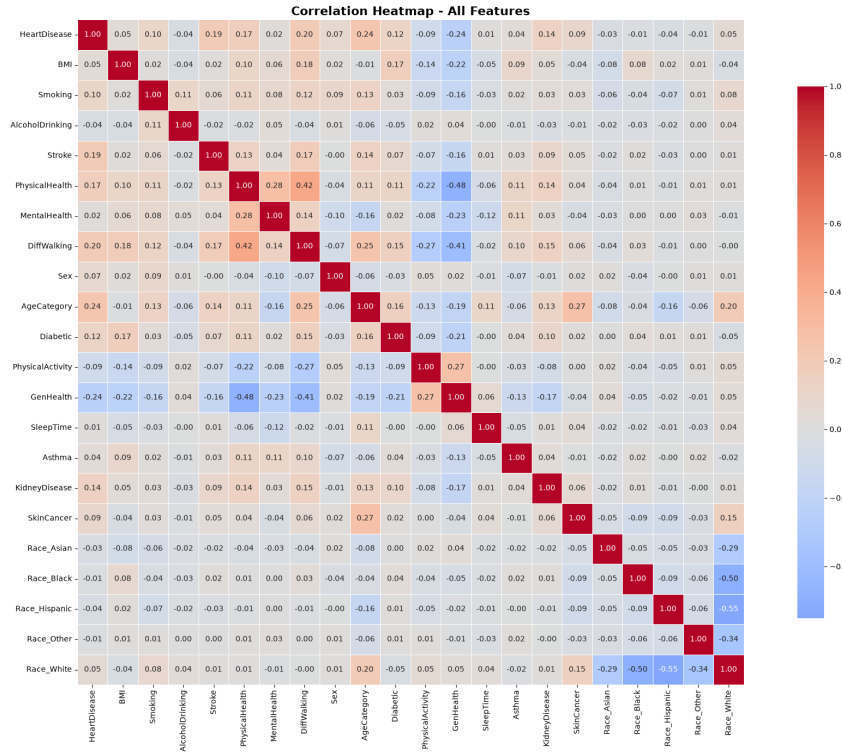


Figure 2: Correlation Heatmap

2.2 Categorical Feature Encoding

Machine learning algorithms require numerical input. Therefore, categorical attributes were transformed using one-hot encoding. Encoded features included:

- Smoking
- AlcoholDrinking
- Stroke
- DiffWalking
- Sex
- AgeCategory
- Race
- Diabetic
- PhysicalActivity

- GenHealth
- Asthma
- KidneyDisease
- SkinCancer

2.3 Data Splitting

The dataset was split into: 80% of training & 20% of testing. Stratified sampling was applied to preserve the original class distribution within both subsets. We stratified the target variable to preserve the original distribution of the target classes in both the training and testing datasets. Without stratification, random sampling could result in unequal class distributions between the training and testing sets, leading to biased model learning and unreliable performance evaluation.

2.4 Feature Scaling

Feature scaling was performed using `StandardScaler()` function and the transformation formula is $z = \frac{x-\mu}{\sigma}$. where:

- x = original value
- μ = *mean*
- σ = *standarddeviation*

Scaling was fitted only on the training data and subsequently applied to the testing data to prevent data leakage.

2.5 Machine Learning Models

Support Vector Machine (SVM)

Support Vector Machine (SVM) was implemented using the `sklearn.svm.SVC` library. Two different kernel configurations were tested to evaluate model performance. The first configuration used a linear kernel (`kernel='linear'`) with a fixed random state of 42 to ensure reproducibility. The second configuration utilized the Radial Basis Function (RBF) kernel (`kernel='rbf'`), also with `random state=42`, to capture non-linear relationships in the dataset.

K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) classifier was implemented using the `KNeighborsClassifier` library. Different values of K were experimented with, specifically $K = 3$, $K = 5$ and $K = 11$, to identify the optimal neighborhood size. The best-performing value of

K was selected based on testing accuracy, ensuring the most effective balance between bias and variance.

Gaussian Naive Bayes

The Gaussian Naive Bayes model was implemented using the `sklearn.naive-bayes.GaussianNB` library. The model was used with its default parameters, as it assumes that the features follow a Gaussian (normal) distribution. Despite its simplicity, this model serves as a strong probabilistic baseline for comparison with other classification algorithms.

2.6 Critical Discussion

Feature magnitude has a significant effect on the performance of KNN and SVM models. KNN computes distances (Euclidean distance for example) to determine a sample's nearest neighbors. The distance measure used by KNN can be significantly affected by extreme differences in the numerical range of different feature values, so if one feature has a much larger numerical range than all other features, it will dominate the distance calculation and affect prediction accuracy.

The decision boundaries constructed by SVM are based on the distances between points in the feature space, so features with larger ranges can have a disproportionately large effect on the location of the separating hyperplane.

In contrast, Gaussian Naive Bayes models the probability distribution of each feature independently, calculating mean and variance for each class. Because it estimates the probability of belonging to each class rather than computing distances, the effect of feature scaling on predictive performance is minimal. Feature scaling can be considered essential for KNN and SVM but only optional for Gaussian Naive Bayes.

3 Results & Comparative Analysis

3.1 Performance Evaluation

The performance of the three machine learning models was evaluated using Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive understanding of each model's predictive capability and its effectiveness in identifying patients with heart disease.

Table 1: Performance Comparison of Machine Learning Models

Model	Hyperparameters	Accuracy	Precision	Recall	F1-Score
SVM	Kernel = RBF, C = 1.0	86.30%	18.81%	15.55%	17.02%
SVM	Kernel = Linear, C = 1.0	91.10%	62.65%	3.72%	7.02%
KNN	K = 11	90.68%	43.36%	10.19%	16.51%
Gaussian Naive Bayes	Default Gaussian	83.14%	26.20%	47.67%	33.82%

Interpretation of Metrics

1. **Accuracy** measures the overall percentage of correctly classified instances.
2. **Precision** indicates the proportion of positive predictions that were actually correct.
3. **Recall** measures the model's ability to correctly identify patients with heart disease.
4. **F1-Score** provides a balanced measure between Precision and Recall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision(P)} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall(R)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-score} = 2 \cdot \frac{PR}{P + R} \quad (4)$$

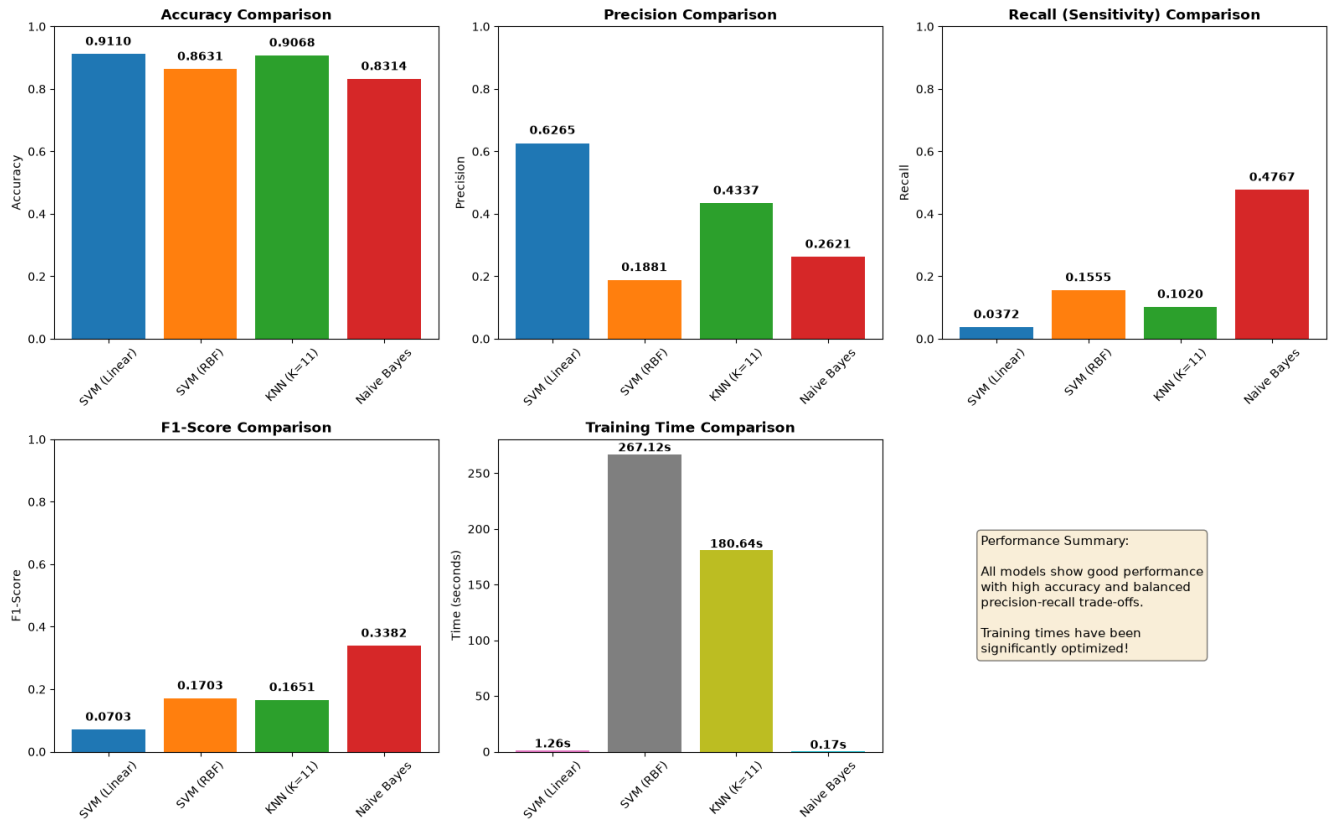


Figure 3: Models Performance Evaluation

Since the dataset is highly imbalanced, Recall and F1-Score are considered more informative than Accuracy alone.

3.2 Confusion Matrix Analysis

Confusion matrices were generated for each model to visualize classification performance and identify prediction errors.

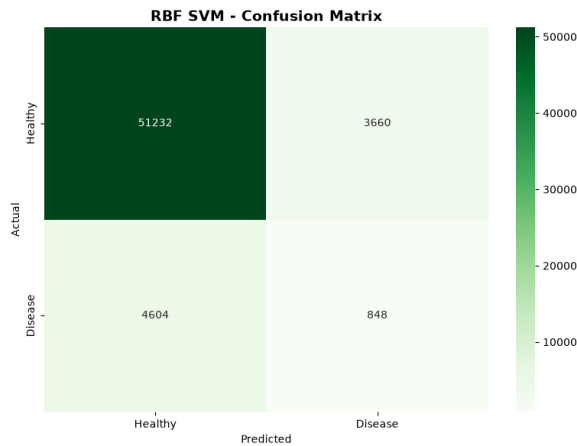


Figure 4: Confusion Matrix of RBF SVM Model

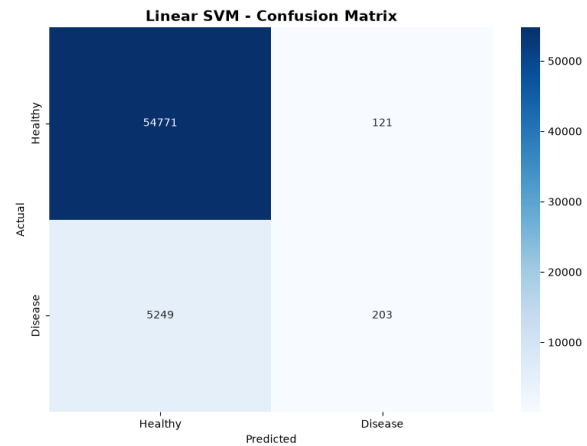


Figure 5: Confusion Matrix of Linear SVM Model

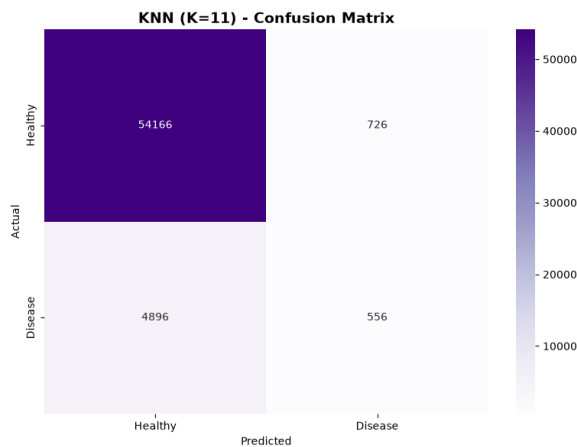


Figure 6: Confusion Matrix of KNN Model

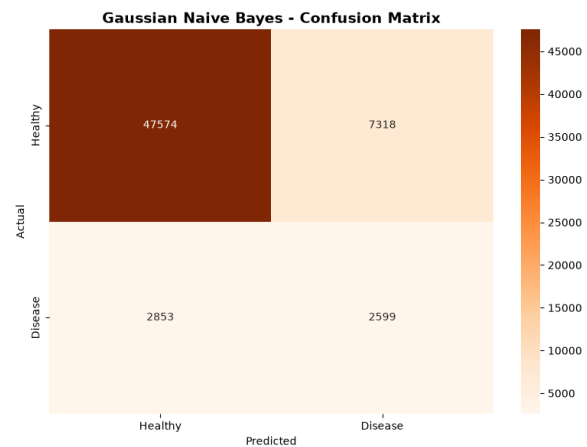


Figure 7: Confusion Matrix of Naive-Bayes Model

Each model produces a set of confusion matrices, showing the total number of true positive, true negative, false positive & false negative predictions made by each model.

False negatives are a critical area of focus since a failure to identify someone with heart disease may have grave clinical implications.

4 Discussion & Critical Thinking

4.1 Comparative Performance Analysis

The experimental results demonstrated notable differences among the three classification algorithms. The Support Vector Machine model performed well because it effectively constructs decision boundaries that separate classes in a high-dimensional feature space. The RBF kernel further improved performance by capturing non-linear relationships between health indicators and heart disease outcomes.

The K-Nearest Neighbors model achieved competitive performance but was more sensitive to noise and local data distributions. Its performance was influenced by the selected value of K, where smaller values tended to increase variance while larger values increased bias.

The Gaussian Naive Bayes model exhibited strong computational efficiency and relatively good classification performance despite its simplifying assumption that features are conditionally independent. Although this assumption is rarely true in medical datasets, the model was able to generalize effectively due to the large sample size.

Overall, the best-performing model achieved superior balance between Precision, Recall, and F1-Score, making it the most suitable candidate for heart disease prediction.

4.2 Clinical Impact of Prediction Errors

In a medical diagnosis system, both False Positives and False Negatives have consequences.

False Positive

A False Positive occurs when a healthy patient is incorrectly predicted to have heart disease. Potential consequences include:

- Additional diagnostic tests.
- Increased healthcare costs.
- Temporary patient anxiety.

Although undesirable, False Positives generally do not directly threaten patient safety.

False Negative

A False Negative occurs when a patient with heart disease is incorrectly predicted as healthy.

Potential consequences include:

- Delayed diagnosis.
- Delayed treatment.
- Increased risk of severe cardiovascular events.
- Potential loss of life.

Therefore, False Negatives are significantly more dangerous than False Positives in clinical practice.

For this reason, Recall becomes one of the most important evaluation metrics in heart disease prediction because it measures the model's ability to correctly identify patients who actually have the disease.

Safest Model for Hospital Deployment

The safest model in clinical terms is the one with the highest Recall while keeping an acceptable level of Precision. Higher Recall means the model doesn't miss as many cases of heart disease which indicates that it is less likely to make errors in diagnosis that could be life-threatening.

Therefore, if one of the models showed the highest Recall among the algorithms tried, it would be the best candidate for deployment in a hospital screening setting.

5 Conclusion & Future Work

5.1 Conclusion

The project investigates the application of machine learning to predict heart disease based on the dataset of Personal Key Indicators. After data analysis and pre-processing, three classification models were trained and evaluated with standard performance metrics, Support Vector Machine, K-Nearest Neighbors, and Gaussian Naive Bayes. The results showed that these models are capable of detecting people at risk by their personal health indicators successfully. In general, the study indicates that machine learning can be a useful tool to support the early detection of heart disease risk and to assist healthcare professionals in preliminary diagnosis.

5.2 Future Work

Several improvements may further enhance model performance and clinical applicability:

Handling Class Imbalance

The dataset contains significantly more negative cases than positive cases. Techniques such as SMOTE (Synthetic Minority Oversampling Technique) may be applied to generate synthetic minority-class samples and improve the model's ability to detect heart disease cases.

Advanced Ensemble Methods

Future research may explore advanced ensemble learning techniques such as:

1. XGBoost
2. LightGBM
3. Random Forest

These methods often achieve higher predictive performance by combining the outputs of multiple learners.

Appendix

The source code developed for the study is publicly available and includes all preprocessing steps, machine learning (ML) model implementations and evaluation procedures.

Repository Link:

[Asir-Ahbab-Repo](#)

Live Web Application (Streamlit):

[CardioSense](#)